RAJALAKSHMI ENGINEERING COLLEGE

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P2

Started on	Thursday, 24 September 2026, 1:43 PM
State	Finished
Completed on	Thursday, 24 September 2026, 1:43 PM
Time taken	39 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1		a	g	**g**	**t**	**a**	**b**
s2		**g**	x	**t**	**x**	**a**	y	**b**

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab azb	2

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
#include <string.h>

int main()
{
    char s1[1000], s2[1000];
    int dp[1000][1000];

    scanf("%s", s1);
    scanf("%s", s2);

    int n = strlen(s1);
    int m = strlen(s2);

    for (int i = 0; i <= n; i++)
    {
        dp[i][0] = 0;
    }

    for (int j = 0; j <= m; j++)
    {
        dp[0][j] = 0;
    }

    for (int i = 1; i <= n; i++)
    {
        for (int j = 1; j <= m; j++)
        {
            if (s1[i - 1] == s2[j - 1])
            {
                dp[i][j] = dp[i - 1][j - 1] + 1;
            }
            else
            {
                if (dp[i - 1][j] > dp[i][j - 1])
```

```
                {
                    dp[i][j] = dp[i - 1][j];
                }
                else
                {
                    dp[i][j] = dp[i][j - 1];
                }
            }
        }
    }

    printf("%d", dp[n][m]);

    return 0;
}
```

	Input	Expected	Got	
✔	aab azb	2	2	✔
✔	ABCD ABCD	4	4	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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